

Semester 1 15 Credits	* Calculus I MATH 1650 4 Credits	Chem for Engineers CHEM 1670 or 1770 & 1770L 4 Credits	Engineering Problem Solving AERE 1600/H 3 Credits	Intro to College Research LIB 1600 1 Credit	* English Composition I ENGL 1500 3 Credits	Engineering Orientation ENGR 1010
Semester 2 15 Credits	* Calculus II MATH 1660 4 Credits	Classical Physics I PHYS 2310 & 2310L 5 Credits	Num., Graph., & Lab Techniques AERE 1610/H 3 Credits	GenEd from Dept List 3 Credits	Aerospace Seminar AERE 1920/H	
Semester 3 18 Credits	Calculus III MATH 2650 4 Credits	Classical Physics II PHYS 2320 & 2320L 5 Credits	Intro to Material Sci & Engr MATE 2730 3 Credits	Statics for Engineers CE 2740 3 Credits	* English Composition II ENGL 2500 3 Credits	
Semester 4 16 Credits	Differential Equations MATH 2670 4 Credits	Dynamics ME 3450 3 Credits	* Intro to Performance & Design AERE 2610 3 Credits	Mechanics of Materials EM 3240 3 Credits	* Technical Comm ENGL 3090 or 3140 3 credits	
Semester 5 17 Credits	Flight Structures AERE 3210 3 Credits	Structures Lab AERE 3220 2 Credits	Thermodynamics ME 2310 3 Credits	Aero-dynamics I AERE 3100 3 Credits	Astro-dynamics I AERE 3510 3 Credits	Flight Dynamics & Control AERE 3550 3 Credits
Semester 6 18 Credits	Advanced Flight Structures AERE 4210 3 Credits	Comp. Techniques for Aero. Design AERE 3610 3 Credits	Aerospace Systems AERE 3620 3 Credits	Aerodynamics/Propulsion Lab AERE 3440 3 Credits	Aero-dynamics II AERE 3110 3 Credits	Flight Control Systems AERE 3310 3 Credits
Semester 7 15 Credits	TechElec from Dept List 3 Credits	Astronautics Requirement 3 Credits (see chart, page 2)	Modern Design Methodology with Aero Applications AERE 4600 3 Credits	OR with Astro Applications AERE 4610 3 Credits	Aerospace Vehicle Propulsion AERE 4110 3 Credits	GenEd from Dept List 3 Credits
Semester 8 15 Credits	TechElec from Dept List 3 Credits	TechElec from Dept List 3 Credits	Design of Systems Aero AERE 4700 OR Space AERE 4620 3 Credits	GenEd from Dept List 3 Credits	GenEd from Dept List 3 Credits	

Course Groups

Basic Program	Aerospace Requirement	Technical Elective	Engineering Fundamentals
Mathematics	Chemistry & Physics	General Education	Seminars

* See back for specific grade requirements

Basic Program - 24 credits

Must be completed (Basic Program GPA $\geq$ 2.0 and Cumulative GPA $\geq$ 2.0) before 200-Level Engineering courses

	Calculus I	MATH 1650	4	Calculus II	MATH 1660	4
	Chemistry	CHEM 1670	4	Physics I	PHYS 2310/2310L	5
	Engr. Prob Solving	AERE 1600/H	3	English Composition I	ENGL 1500	3
	Engr. Orientation	ENGR 1010	R	Intro to College Research LIB	1600	1

English Proficiency - 9 credits

	English Comp I	ENGL 1500	3
	English Comp II	ENGL 2500	3
	Technical Comm	ENGL 3090/3140	3

Grade of C or higher is required in ENGL 1500, ENGL 2500 and ENGL 3090/3140.

General Education - 12 credits

	GenEd
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Visit <https://www.aere.iastate.edu/forms/> for the approved GenEd List.

Must include U.S. Cultures and Communities (3 credits) and International Perspective (3 credits) requirement.

Two semester sequence in a single world language may be applied per ISU World Language Requirements.

Aerospace Engineering - 50 credits in 8 areas of study

	Aerodynamics	Aerodynamics I - Incompressible Flow Aerodynamics II - Compressible Flow Aerodynamics/Propulsion Lab	AERE 3100 AERE 3110 AERE 3440	3 3 3
	Propulsion	Aerospace Vehicle Propulsion	AERE 4110	3
	Structures	Flight Structures Aero Structures Lab Adv Flight Structures	AERE 3210 AERE 3220 AERE 4210	3 2 3
	Controls	Flight Dynamics & Control Flight Control Systems	AERE 3550 AERE 3310	3 3
	Astrodynamics	Astrodynamics I	AERE 3510	3
	Systems and Design	Performance and Design Aerospace Systems Design Methodology Design of Aero Systems	AERE 2610 AERE 3620 AERE 4610 AERE 4620	3 3 3 3
	Software and Numerics	Num., Graph., & Lab Techniques Comp. Techniques for Aero. Design	AERE 1610/H AERE 3610	3 3
	Astronautics Requirement	One of: (a) Rocket Propulsion (b) Spacecraft Dynamics	AERE 4150 AERE 4330	3 3

Technical Electives - 9 credits

	Make to Innovate/ Undergrad Research	AERE 2940 - Freshman and Sophomores (credits do not count towards graduation). AERE 4940 - Juniors and Seniors (maximum of 6 credits may count towards graduation).
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	Technical Electives	Group A Aerospace Electives (3 credits), Group B Technical/Engineering Electives (3 credits) and Group C Career Electives (3 credits). Group A electives can be used for Group B and C. Group B electives can be used for Group C. Visit https://www.aere.iastate.edu/forms/ for the approved Technical Electives List.
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	Grade Requirements for Courses	Grade of C or higher is required in ENGL 1500, 2500, and 3090/3140. Grade of C- or higher in MATH 1650 and 1660. Grade of C- or higher in AERE 2610.
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